

SPSS 5 Linear Regression

01/07/2015

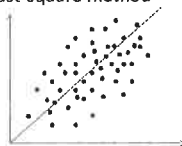
Topics

- Regression analysis
- Multiple regression and R^2
- Automatic model building

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Regression Analysis

- Regression Analysis
 - Use **correlation** to construct a **regression line** model (equation) of 2 variables
 - Use the regression line to **predict** a variable given another variable
- Regression Line
 - The prediction line that lies closest to all samples in the scatterplot
 - Calculated using least-square method



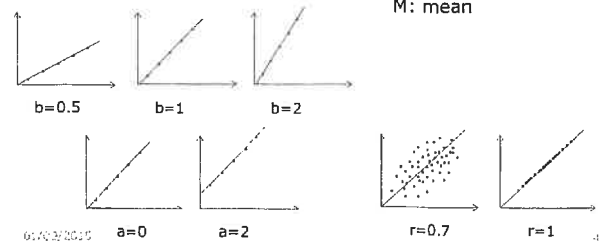
Regression Line Components

- Equation of Regression Line:
 - b : Slope
 - a : Point of intercept
 - r : Extend of cluster around line

$$Y_{pred} = a + bX$$

$$b = \frac{r s_y}{s_x}, a = M_y - b M_x$$

where
 s : standard deviation,
 M : mean



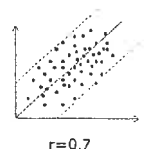
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Regression Line Components (con't)

- Correlation (r value, Pearson r):
 - Scatter around line
 - r^2 : The proportion of y variation influenced by x variation
 - $1-r^2$: the proportion of y variation that is random
- Beta coefficient (b value):
 - Slope of the regression line, 45° line for $b=1$
 - The ratio of change per unit for y over x
- Intercept (a value):
 - Point of y -intercept

Uncertainty of prediction

- Observation:
 - All samples scattering around the regression line within a range
 - The pattern is partly caused by the **correlation (straight line)**, and partly **random** (normal distribution)
 - Because of the randomness, the prediction based on the regression line is a little **uncertain**
- How uncertain?



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Check Regression Assumptions

- Residuals are the distances between individual samples and the regression line
- Use residuals to check regression assumptions
- Types of residuals:
 - Unstandardized residuals
 - Difference between observed and predicted values
 - Standardized residuals
 - Transformed residuals that have SD of 1
 - Studentized residuals
 - Transformed residuals divided by its estimated SD
 - Deleted residuals
 - Identify influential cases that once deleted, increases the residuals
 - Studentized deleted residuals
 - Deleted residuals divided by their SD

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Check Regression Assumptions

- To obtain Studentized residuals:
 - Recall the Linear Regression Dialogue
 - Save > Residuals > select Studentized > Continue > OK

A variable SRE_1 of Studentized residue is added for each sample.
- To obtain predicted values:
 - Recall the Linear Regression Dialogue
 - Save > Predicted Values > select Unstandardized > Continue > OK

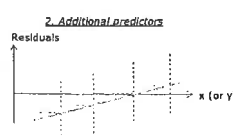
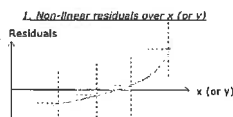
A variable PRE_1 of unstandardized value is added for each sample.

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Check Regression Assumptions

- Check linearity
 - Check scatter plot for points clustering around a line
 - Plot residuals against independent or predicted variable. It should be a random scatter around 0
- Identify additional predictors
 - Plot the residuals against values of independent variables that are NOT in the equation
 - If a relationship between the variables exist, add the variable in the regression model

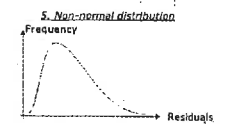
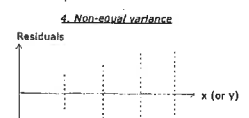
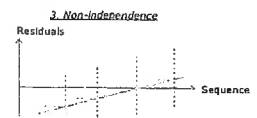


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Check Regression Assumptions (Con't)

- Check independence
 - Check whether time sequence of the cases have relationship with the residuals.
 - Plot the residuals against the order in which they were obtained
- Check equal variance
 - Plot residuals against predicted variable. See if the spread of residuals changes.
- Check normal distribution
 - Make histograms or stem-and-leaf plots of the residuals to check normal distribution



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Task - Check Regression Assumption

- Check linearity and equal variance by plotting residuals (SRE_1) against predicted variable (PRE_1)
- Check independence with plot of residuals against the data order (caseno)
- Check normal distribution of residual with histogram of residuals (SRE_1)

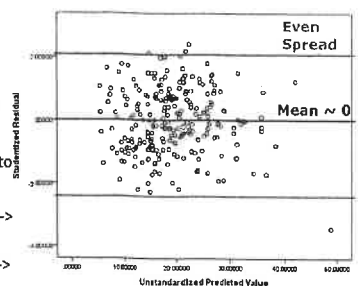
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Task Answer- Check Regression Assumption

Check linearity and equal variance:

- Graphs > Legacy Dialogs> Scatter/Dot > Simple Scatter > Define
 - Set Y Axis to SRE_1, set X Axis to PRE_1 > OK
- The residuals scatters around 0 -> The relationship seems linear
 - The spread of residuals is even -> The variances appear equal



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Task: Multiple Regression Coefficients

- Multiple regression coefficients
 - Analyze > Regression > Linear
 - Set **Dependent** to fatpct1
 - Set **Independent** to abdomcirc, hipcirc, wristcirc
 - OK

Model Summary^a

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.648 ^a	.722	.718	4.44925

a. Predictors: (Constant), Wrist circumference (cm), Abdominal circumference (cm), Hip circumference (cm)
b. Dependent Variable: Percent fat method 1

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error			
1	(Constant)	1.551	6.712		.250	.603
	Abdominal circumference (cm)	.928	.055	1.195	17.026	.000
	Hip circumference (cm)	-.329	.063	-.292	-3.979	.000
	Wrist circumference (cm)	-1.944	.384	-.217	-4.922	.000

a. Dependent Variable: Percent fat method 1

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Task: Multiple Regression Coefficients

- Model summary table:
 - R square improve to .722 compared with .563 with using abdomcirc only
 - 3 variables explain 72.2% variability in body fat
- Coefficients table
 - Positive correlation** for abdominal circumference
 - Negative correlation** for hip and wrist circumference

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a. Dependent Variable: Percent fat method 1

$$\text{PredFat} = 1.551 + (0.928 \times \text{AbdomCirc}) + (-0.329 \times \text{HipCirc}) + (-1.944 \times \text{WristCirc})$$

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More about Multiple R Regression

- R value
 - The multiple R should only be used for prediction when it is higher than all the internal correlations with y, e.g. $R_{y,1,2} > r_{y,1}, r_{y,2}, r_{1,2}$
 - Otherwise, use internal correlations instead to predict
- ANOVA: test for R significant
 - Convert R to F and test F with critical F value for significance (You will learn about F in ANOVA. In general, high F with small sig means better correlation)

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More about Multiple R

- Limit to Multiple R
 - Multiple R is best to be used when predictors are orthogonal (independent), not correlated
 - If two predictors are dependent, one of them will be close to 0 due to duplicated effect.

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Automatic Model Building

- Sequentially add or remove independent variables from the model to build a model
- Need not to evaluate all possible combinations
- Based on the change in R^2 after addition/removal of a variable
- Common methods of variable selection
 - Forward selection
 - Backward selection
 - Stepwise selection

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Stepwise Variable Selection

- Most common method
- Sequence:
 - Add variable with the largest increase in R^2 , which is large enough to reject null hypothesis
 - Remove the variable changes R^2 least, if the change is small enough to accept null hypothesis
 - Repeat the sequence until no more variable can be added while satisfying the R^2 criterion

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